

HACKATHON PROJECT REPORT

AIVI

Integrated Disaster Warning System

Borneo Hackweekend 2026 • UNIMAS, Malaysia

Team Name	Garuda Stack
Event	Borneo Hackweekend 2026
Organizer	UNIMAS, Malaysia
Category	Disaster resilience AI
Platform	Web Application (React + Vite)
Deployment	Cloud-based — Vercel

01

Problem Statement

Indonesia sits at the intersection of three major tectonic plates and lies within the Pacific Ring of Fire, making it one of the most disaster-prone nations on Earth. Annually, the archipelago faces flooding, earthquakes, volcanic eruptions, tsunamis, and landslides — events that collectively affect millions of people across thousands of islands.

Despite advances in monitoring technology by agencies such as BMKG (Meteorology, Climatology, and Geophysics Agency) and BNPB (National Disaster Management Authority), a persistent and critical gap remains between the moment anomalies are detected and the moment affected communities receive actionable information. This gap — information asymmetry — is the root cause of preventable casualties and economic losses.

Core Problem: Information latency during a disaster directly amplifies its impact. Each minute of delay translates to missed evacuation windows, compounded panic, and increased loss of life.

KEY PAIN POINTS IDENTIFIED

- Early warning data from central monitoring agencies is available but fails to reach the public in a timely, digestible format.
- Existing communication channels (television, radio, SMS broadcast) are unidirectional and lack geographic precision or user personalization.
- Citizens in affected zones have no structured channel to report ground-level conditions, leaving rescue teams without real-time situational data.
- Mitigation literacy remains low; people lack access to consolidated, location-specific evacuation guidance at the moment they need it most.
- Disaster alert systems are largely reactive — they inform after the fact rather than proactively guiding users through preparation and response.

02 Objectives & UN-SDG Alignment

AIVI is guided by three specific, measurable, and actionable objectives that directly correspond to identified gaps in the current disaster response ecosystem:

1. **To provide real-time, location-personalized disaster alerts** that notify users based on their registered geographic location, eliminating the delay between data acquisition and public communication.
2. **To deliver AI-powered, contextual mitigation guidance** that goes beyond generic alerts — providing users with specific, actionable steps based on the type, severity, and proximity of a disaster event.
3. **To enable community-driven situational reporting** so that survivors and witnesses in affected areas can submit real-time ground-truth data, creating a feedback loop that improves search and rescue (SAR) operations.

UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS (UN-SDG)

AIVI explicitly addresses the following UN-SDG goals through its core functionality:

SDG 3 Good Health & Well-Being	SDG 11 Sustainable Cities & Communities	SDG 13 Climate Action	SDG 17 Partnerships for the Goals
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SDG 3 — Good Health & Well-Being	Reducing disaster-related casualties and injuries by ensuring timely evacuation alerts, emergency contact directories, and immediate AI-generated health and safety guidance reach affected citizens before and during crisis events.
SDG 11 — Sustainable Cities & Communities	Strengthening urban and rural disaster resilience by providing infrastructure-level early warning systems accessible to all citizens

	regardless of technical literacy. AIVI supports local governments in building communities that are safer and more prepared.
SDG 13 — Climate Action	As climate change intensifies the frequency and severity of natural disasters in Southeast Asia, AIVI serves as a direct adaptive response tool. Its architecture is designed to integrate with IoT sensor networks and climate data APIs for long-term climate risk monitoring.
SDG 17 — Partnerships for the Goals	AIVI is architecturally designed to connect with public APIs from authorized bodies (BMKG, BNPB, BPBD), establishing a validated data ecosystem. It models the kind of public-private data partnership essential for scalable, credible disaster management.

03 Proposed Solution

AIVI (Integrated Disaster Warning System) is a cloud-based, AI-augmented public information platform designed to close the information asymmetry gap in disaster response. It operates as a proactive command center — not merely a news feed — that actively matches disaster events to affected users and delivers contextually relevant guidance in real time.

Core Value Proposition: The right information, to the right person, at the right time — automatically determined by geographic proximity and disaster severity.

HOW AIVI WORKS — FOUR FUNCTIONAL PILLARS

01 Real-Time Monitoring Dashboard

An interactive web interface that visualizes active disasters across Indonesia as geographic polygon overlays on a dark-themed map. Severity is color-coded (Critical = Red, High = Orange, Medium = Yellow, Low = Green). Users can filter by disaster type, province, and status. Data auto-refreshes every 30 seconds and supports live WebSocket updates for instantaneous event propagation.

02 Tiered Early Warning Broadcast

A location-aware alert engine that categorizes each user's exposure level as DIRECT (area within the disaster polygon), NEARBY (surrounding affected zone), or INFORMATIONAL (general awareness). Each tier triggers a distinct UX: a full-screen blocking modal with forced acknowledgment for critical threats, a dismissible banner for nearby warnings, and a subtle toast notification for general information.

03 Literacy & Mitigation Center

Each disaster entry is enriched with structured mitigation data: evacuation instructions, nearest assembly point routes, and emergency contact directories for relevant

authorities (BPBD, PMI, Basarnas, BMKG). This turns AIVI into an actionable reference — not just an alert system — that educates users on correct emergency response behavior before, during, and after a disaster.

04 Community-Based Reporting (Crowdsourcing)

A two-way information channel that allows survivors and eyewitnesses in affected zones to submit ground-level field reports. This crowdsourced data layer augments official data with real-time situational intelligence, providing Search and Rescue (SAR) teams with a more accurate operational picture and improving the overall quality of information available to both responders and the public.

TECHNOLOGY STACK

AIVI is engineered on a modern, component-based frontend architecture with a clearly defined integration interface for backend services:

Layer	Technology	Rationale
Frontend Framework	React 18 + Vite	Reactive, component-based architecture ensuring smooth UX across devices
Styling	Tailwind CSS v4	Utility-first responsive design with custom dark-theme design tokens
State Management	Zustand	Lightweight global state with localStorage persistence for session continuity
Map Engine	React Leaflet	Interactive polygon-based geospatial visualization of affected areas
Real-time Layer	Socket.IO Client	WebSocket connection for live disaster event propagation from backend
HTTP Client	Axios	Interceptor-configured REST client with JWT token injection
Form Validation	React Hook Form + Zod	Schema-based validation for registration and location selection
Deployment	Vercel (Cloud)	Edge-deployed for high availability and CDN performance under traffic spikes

SUSTAINABILITY & SCALABILITY

AIVI is intentionally designed to be future-proof. The current prototype establishes the foundation for a production-grade disaster management platform through the following architectural decisions:

- Modular API Integration Layer — the service layer (api.js) is structured to connect with official public APIs from BMKG, BNPB, and regional BPBD offices, ensuring data validity and traceability from authoritative sources.
- IoT Sensor Readiness — the real-time event pipeline (useRealtimeDisasters hook + socketService) is pre-architected to receive automated data feeds from IoT field sensors, enabling fully automated disaster detection without human data entry.

- **Cloud-Native Deployment** — hosted on Vercel with edge distribution, AIVI is designed to remain stable under the extreme traffic spikes that inevitably occur in the early moments of a major disaster.
- **AI Expansion Path** — the AI Recommendation Panel is designed as a standalone service integration point, allowing the current prompt-based model to evolve toward predictive risk modeling and pattern recognition over historical disaster data.
- **Open Architecture** — all state, services, and components follow separation-of-concerns principles, making the codebase maintainable, testable, and ready for contribution from future developers.

04 AI Acknowledgement

In the spirit of transparency and academic integrity, this section provides a full disclosure of all Artificial Intelligence tools used during the development of AIVI. The team acknowledges that AI tools were utilized as development accelerators and ideation aids, not as replacements for original technical work and design decisions.

Disclosure Statement: All AI-generated code, content, and suggestions were reviewed, validated, and adapted by the team. Final architectural decisions, system design, and problem framing are the original intellectual work of the AIVI team.

AI TOOLS USED

AI Tool	Role in Project
Claude (Anthropic)	AI-powered situational recommendations per disaster event; generating personalized mitigation guidance for affected users.
GitHub Copilot	Development assistance during coding; autocomplete suggestions and boilerplate generation for React components and utility functions.
ChatGPT (OpenAI)	Brainstorming feature ideas, drafting documentation, and reviewing report structure during the planning phase.
Cursor IDE	AI-assisted code editor used throughout the development process for faster iteration and refactoring.

SCOPE OF AI USAGE

The following areas of the project involved AI tool assistance:

- Frontend component scaffolding and Tailwind CSS styling patterns (GitHub Copilot)
- Prompt engineering for the AI Recommendation Panel — defining how Claude generates disaster-specific situational guidance (Claude, Anthropic)
- Report writing structure and English language review (ChatGPT)
- Code refactoring suggestions and React best practice guidance (Cursor IDE + Copilot)

The following were not AI-generated and represent the team's original work:

- System architecture design (tiered alert model, impact level classification, component hierarchy)
- Data model design for disasters, alerts, and user locations
- UX flow design including the three-tier alert system (Critical Modal, Warning Banner, Info Toast)
- Problem framing, SDG alignment rationale, and solution methodology

AIVI — Integrated Disaster Warning System

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